Division **E**

# Mathematical Olympiads for Elementary & Middle Schools

**November 2024**

Contest **1**

*Directions to Students: After all questions have been read by your PICO, you will have 30 minutes to complete this contest. You may not have a pen or pencil in your hand while the PICO reads the set of questions to the class. Calculators are not permitted. Answers must be placed in the corresponding boxes in the answer column.*

*Name:* _____

**1A** Compute the product: $2 \times 4 \times 5 \times 15 \times 25$.

**1B** Jacob has a lucky number, and Emma wants to know it. Jacob tells Emma that if she takes his lucky number, triples it, and adds 95, the result will be 2024. What is Jacob's lucky number?

**1C** Emily is playing the video game Super Marlo Sisters. During her game, she collects 8 boxes, each of which has either a coin (worth 100 points) or a flower (worth 250 points). The total score from the 8 boxes is 1250 points. How many of Emily's boxes have flowers?

| Answer Column |
|---|
| **1A** |
| **1B** |
| **1C** |
| **1D** |
| **1E** **in.** |
| *Do Not Write in this Space. For PICO's Use Only.* *SCORE:* |

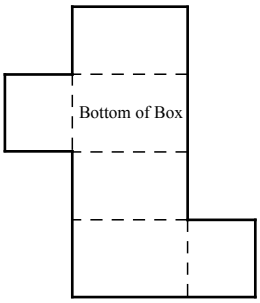
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*Name:* _____

**1D** Nicole has \$4.95 in quarters and dimes. If the dimes were quarters and the quarters were dimes, she would have \$0.45 less. How many quarters does Nicole have?

**1E** A cardboard box with a lid has a volume of 672 $in^3$. The dimensions of the bottom of the box are 8 inches and 12 inches. The box is unfolded to form a flat piece of cardboard, as seen in the diagram.
Calculate the number of inches in the perimeter of the decagon (10-sided shape).

[The diagram is drawn to scale.]



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# SOLUTIONS AND ANSWERS

**1A** **METHOD 1** *Strategy: Arrange the factors to find products that end in 0.*
Regroup the factors as $(2 \times 5) \times (4 \times 25) \times 15 = 10 \times 100 \times 15 =$ **15,000**.

**METHOD 2** *Strategy: Use factorization to simplify the arithmetic.*
Factor each number into primes: $2 \times 2^2 \times 5 \times 3 \times 5 \times 5^2 = 2^3 \times 3 \times 5^3 \times 5 = (2 \times 5)^3 \times (3 \times 5) = 1000 \times 15 = 15{,}000$.

*FOLLOW UP: Compute the product $2 \times 8 \times 5 \times 15 \times 25$.* [30,000]

**1B** **METHOD 1** *Strategy: Work backwards.*
Working backward requires one to use the opposite operations:
Since 2024 – 95 = 1929, and 1929 ÷ 3 = 643, Jacob's lucky number is **643**.

**METHOD 2** *Strategy: Represent the lucky number as a variable and use algebra to write and solve an equation.*
Let $c$ represent the lucky number. Then, $3c + 95 = 2024$. Solve this equation for $c$ by subtracting 95 and then dividing by 3: $3c + 95 = 2024 \rightarrow 3c = 1929 \rightarrow c = 643$.

*FOLLOW UP: If you add 56 to the year Mary was born and divide the sum by 25, the result is 80. What year was Mary born?* [1944]

**1C** **METHOD 1** *Strategy: Make a supposition.*
Suppose each of the 8 boxes is worth 100 points. This gives a total of $8 \times 100 = 800$ points, which is 450 points too low. Changing a coin box (100 pts) to a flower box (250 pts) increases the total by 150 points. We need to change 450/150 = **3** coin boxes to flower boxes. Confirm that 5(100) + 3(250) = 1250 points.

**METHOD 2** *Strategy: Use algebra to write a system of equations.*
Let $c$ represent the number of coin boxes, each worth 100 points, and let $f$ represent the number of flower boxes, each worth 250 points. Use the equations $c + f = 8$ and $100c + 250f = 1250$ to find $f$. Rewrite the first equation as $c = 8 - f$ and substitute this into the second equation: $100(8 - f) + 250f = 1250$. Thus, $800 - 100f + 250f = 1250 \rightarrow 800 + 150f = 1250 \rightarrow 150f = 450$, so $f = 3$.

**METHOD 3** *Strategy: Create a table of possible values.*

| Coin | 8 | 7 | 6 | **5** |
|---|---|---|---|---|
| Flower | 0 | 1 | 2 | **3** |
| Total | 800 | 950 | 1100 | **1250** |

The table shows that 5 boxes containing coins and 3 containing flowers results in 1250 points.

*FOLLOW UP: At the Super Cup, Kaden buys keychains and hats for his friends. Each keychain cost \$10 and each hat \$25. If Kaden bought 8 items and spent \$125, how many keychains did he buy?* [5]

| Answer Column |
|---|
| 1A **15,000** |
| 1B **643** |
| 1C **3** |
| 1D **15** |
| 1E **112 in.** |

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**1D** **METHOD 1** *Strategy: Use logic and create tables reflecting possible answers.*
Begin by maximizing the number of quarters (19) and then adding dimes (2) to make \$4.95, as shown in the table below. Then reduce the number of quarters by 2 and test the value of the reversed coin count. Repeat until the reversed coin value equals \$4.95 – \$0.45 = \$4.50.

| Quarters | 19 | 17 | 15 |
|---|---|---|---|
| Dimes | 2 | 7 | 12 |
| Quarters | 2 | 7 | 12 |
| Dimes | 19 | 17 | 15 |
| Value | $0.50 + 1.90 \neq \$4.50$ | $1.75 + 1.70 \neq \$4.50$ | $3.00 + 1.50 =$ **\$4.50** |

Nicole has **15** quarters.

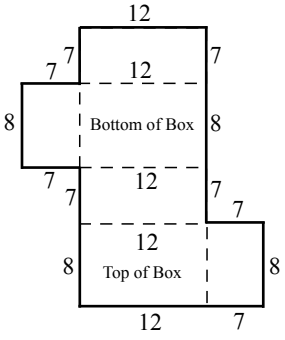
**METHOD 2** *Strategy: Eliminate possibilities using mathematical reasoning.*
There must be an odd number of quarters to have a total of \$4.95. We also know that 4.95 – 0.45 = \$4.50. Since 17 quarters + 7 dimes = \$4.95 and 17 dimes + 7 quarters = \$3.45, we need to reduce the number of quarters. The next possible odd number of quarters would result in 15 quarters + 12 dimes = \$4.95. Since 15 dimes + 12 quarters = 1.50 + 3.00 = \$4.50, which is exactly \$0.45 less than \$4.95, we know Nicole has 15 quarters.

**METHOD 3** *Strategy: Consider the change in value of swapping one dime with one quarter.*
The difference in value between a dime and a quarter is \$0.15. Since 0.45/0.15 = 3, the number of dimes and quarters must differ by 3. Let $d$ = the number of dimes and $d + 3$ = the number of quarters. Then $0.10d + 0.25(d + 3) = 4.95 \rightarrow 0.10d + 0.25d + 0.75 = 4.95 \rightarrow 0.35d = 4.20 \rightarrow d = 12$. Thus, there are 12 dimes and 15 quarters.

*FOLLOW UP: Aaron has \$4.00 in quarters and nickels. If the nickels were quarters and the quarters were nickels, he would have half as much money. How many quarters does Aaron have?* [15]

**1E** **METHOD 1** *Strategy: Find the height of the box, label the diagram, and add the side lengths.*
To find the perimeter, first find the height of the box. Volume = $l \times w \times h = 12 \times 8 \times h = 96h$. Since the volume is 672 in$^3$, we solve $96h = 672$ to get $h = 7$.
The perimeter of the decagon = 2(12) + 4(8) + 8(7) = 24 + 32 + 56 = **112** in.

**METHOD 2** *Strategy: Subtract the measures of the folded edges from the total perimeter of the 6 rectangles.*
Find the height of the box as in Method 1. The top and bottom rectangles are 8 by 12 and each have a perimeter of 40 in. The front and back rectangles are 12 by 7 and each have a perimeter of 38 in. The left and right rectangles are 7 by 8 and each have a perimeter of 30 in. The total perimeter of the 6 rectangles = 2(40 + 38 + 30) = 216 in. There are 5 "folded" edges of the box, indicated by the dotted lines in the diagram. Since each one is counted twice in the perimeters of the 6 rectangles, the total of those edges is 2(12 + 12 + 12 + 8 + 8) = 104 in. Thus, the perimeter of the decagon is 216 – 104 = 112 in.




*FOLLOW UP: What is the surface area of the cardboard box in the problem?* [472 in$^2$]

***NOTE:*** *Other FOLLOW UP problems related to some of the above can be found in our four contest problem books and in "Creative Problem Solving in School Mathematics." Visit www.moems.org for details and to order.*